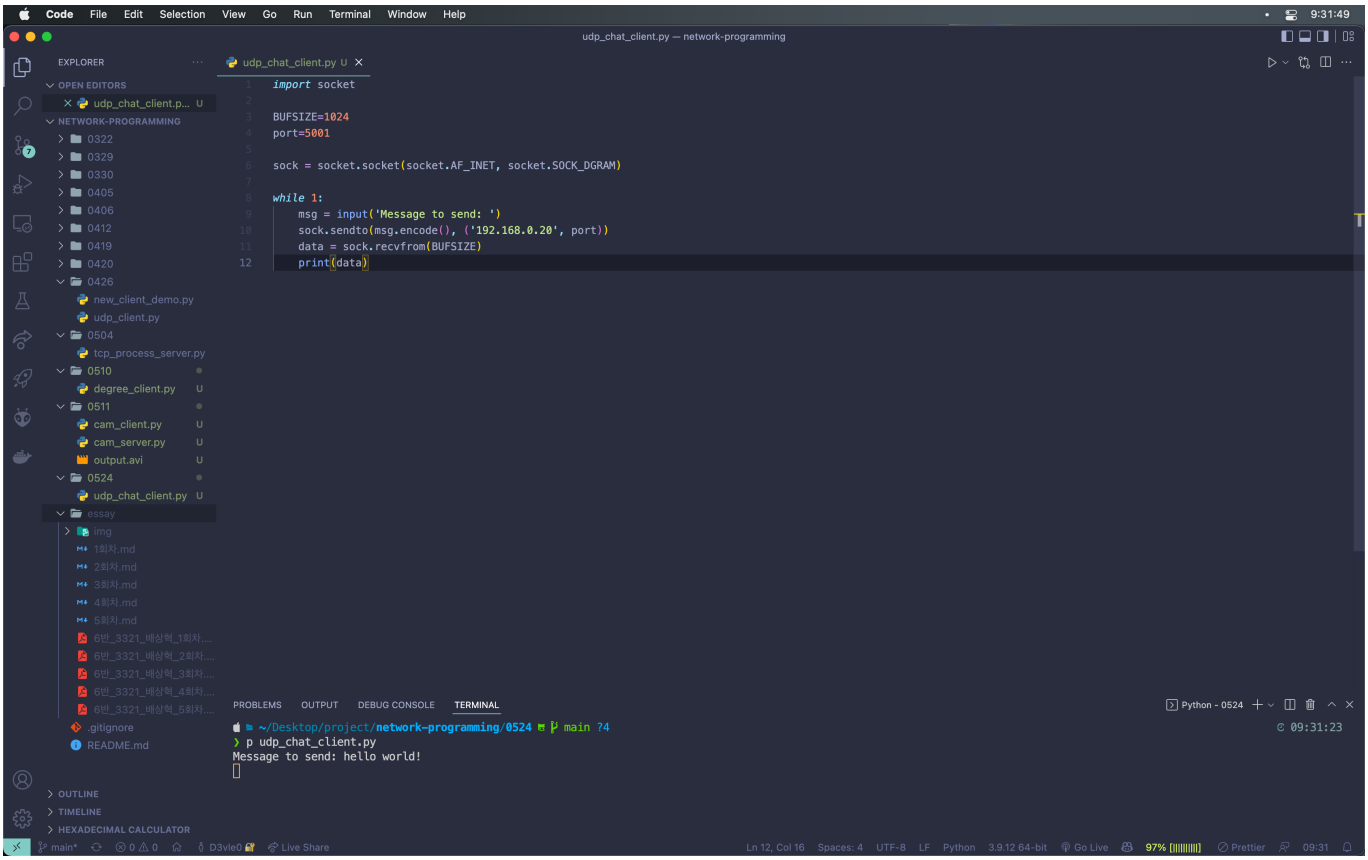
6반_3321_배상혁_6회차

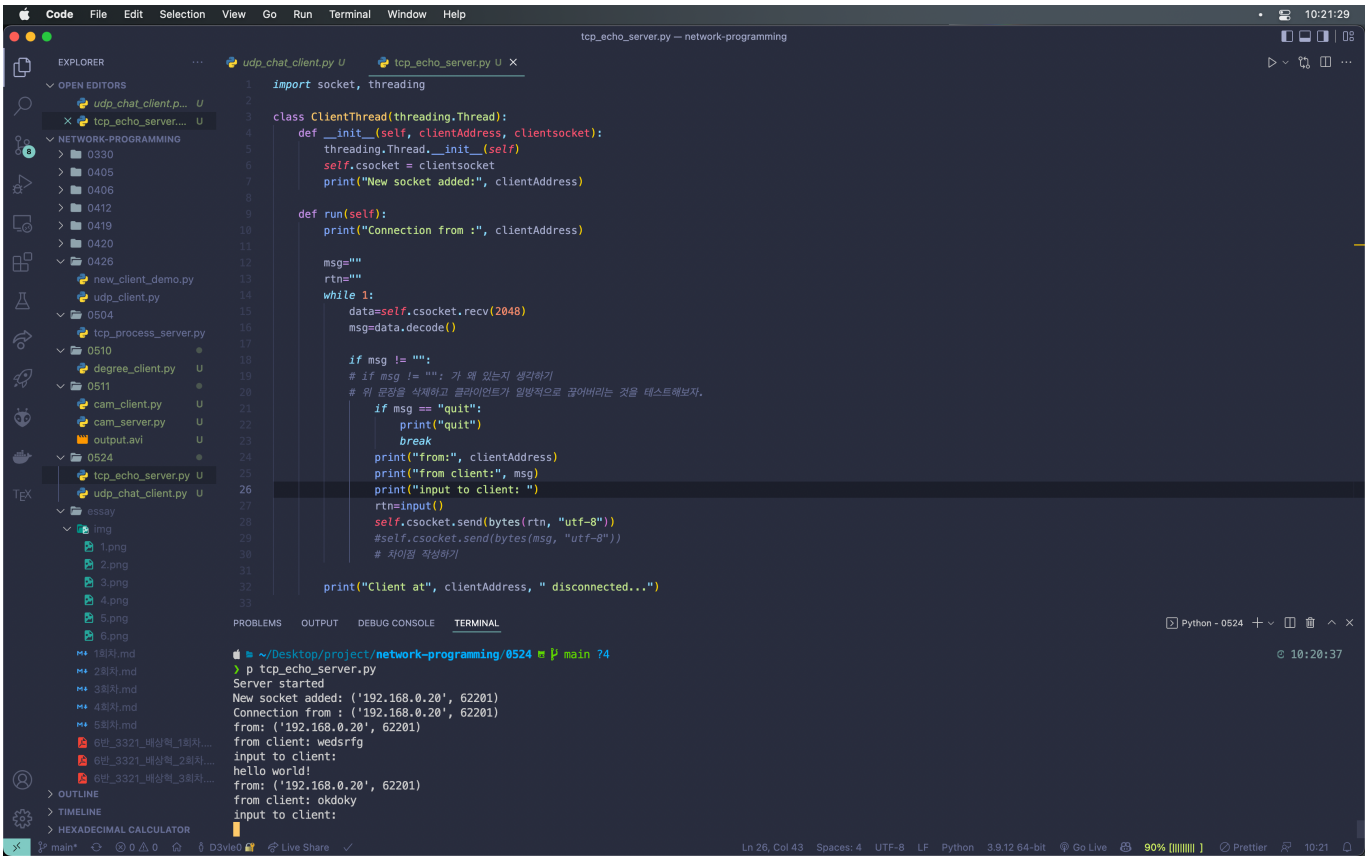



```python
# 6회차
import socket

BUFSIZE=1024
port=5001

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

while 1:
    msg = input('Message to send: ')
    sock.sendto(msg.encode(), ('192.168.0.20', port))
    data = sock.recvfrom(BUFSIZE)
    print(data)
```




```python
# 6회차
import socket, threading

class ClientThread(threading.Thread):
    def __init__(self, clientAddress, clientsocket):
        threading.Thread.__init__(self)
        self.csocket = clientsocket
        print("New socket added:", clientAddress)

    def run(self):
        print("Connection from :", clientAddress)

        msg=""
        rtn=""
        while 1:
            data=self.csocket.recv(2048)
            msg=data.decode()

            if msg != "":
                if msg == "quit":
                    print("quit")
                    break
                print("from:", clientAddress)
                print("from client:", msg)
                print("input to client: ")
                rtn=input()
                self.csocket.send(bytes(rtn, "utf-8"))
                #self.csocket.send(bytes(msg, "utf-8"))
```

```
            print("Client at", clientAddress, " disconnected...")

LOCALHOST = "192.168.0.33"
PORT = 5001

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
server.bind((LOCALHOST, PORT))
print("Server started")

while 1:
    server.listen(5)
    clientsock, clientAddress = server.accept()
    newthread = ClientThread(clientAddress, clientsock)
    newthread.start()
```

Assignment 1.

> Difference between the existence of `if msg != "":`

I tested that the client disconnects the connection unilaterally depending on the presence or absence of `if msg != "":`. If there is `if msg != ""`, a BrokenPipeError is raised in the code that sends the message. According to the official Python documentation, this occurs when an attempt is made to write to a pipe that has the other end closed, or when a write is attempted to a socket that has ended. Simply put, this happens when a process is trying to write data to a pipe, but there is no other process accepting it. If there is no `if msg != ""` , a ConnectionResetError is raised in the code receiving the message. According to the official Python documentation, this happens when a connection is forcibly terminated by the other party.

Assignment 2.

> Difference between `self.csocket.send(bytes(rtn, "utf-8"))` and `self.csocket.send(bytes(msg, "utf-8"))`

The variable rtn is the message the server sends to the client, and the variable msg is the message the server receives from the client. Therefore, if you send an rtn to the client, the content the server wants to send is normally sent, but if you send an msg, the message received from the client is sent, and as a result, the client receives the message sent by itself.